2nd Wind Conclave

FOH CHANNEL EQ SETUP — DiGiCo Quantum 225

Show	2nd Wind Conclave
Show Date	2026-07-31
Venue	Fountain Square (outdoor)
Console	DiGiCo Quantum 225
Channels	27
FOH Engineer	Brian Lloyd
Monitor Engineer	
Show Time	6:00pm–11:00pm
Rev	Rev 1.0 Deep Build

DOCUMENT CONTENTS

1. Cover Page — show, venue, and personnel
2. Input List — color-coded full channel patch sheet
3. Patching — AES and Local inputs sorted by port
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5. Reference — EQ structure, pan guide, bus grouping, soundcheck order

SHOW STYLE: Omega Psi Phi 85th Grand Conclave block party — a 10,000-person dance floor on an open plaza. R&B;, funk, Motown and Top 40 from a four-vocalist show band running to a click with tracks under it.

2nd Wind Conclave

VENUE: Fountain Square (outdoor)	**DATE: 2026-07-31**	**REV: Rev 1.0 Deep Build**
FOH: Brian Lloyd	**MON:**	**SHOW TIME: 6:00pm–11:00pm**

Ch	Instrument	Mic / DI	Patch	48V	Stand	Notes
■ DRUMS						
1	Kick In	Shure Beta 91A	Local 1	✓	—	Boundary mic on the pillow, aimed at the beater. Contour switch assumed OUT (flat).
2	Kick Out	Audix D6	Local 2		Short	Outside the resonant head at the port. Polarity check against ch 1 at soundcheck.
3	Snare Top	Audix i5	Local 3		Short	LOCKER SWAP — i5 from the DP8, replacing the Beta 98H/C. Dynamic, so no 48V.
4	Snare 2	Shure Beta 98H/C	Local 4	✓	Clip	Rim-mounted gooseneck. CONFIRM AT LOAD-IN: built as a second snare drum, not a bottom head. If it's the bottom, polarity flip and a different curve.
5	Hat	Electro-Voice N/D 408	Local 5		Boom	Confirmed by Brian as the EV N/D 408 (not the Lauten LS-408). Windscreen — gusts to 16 mph.
6	Rack 1	Audix D2	Local 6		Clip	Template ships this fader gated — thr –36.2 dB, rel 227 ms, 130–317 Hz sidechain. Left as-is.
7	Rack 2	Audix D2	Local 7		Clip	Template ships this fader gated — thr –36.2 dB, rel 227 ms, 130–317 Hz sidechain. Left as-is.
8	Floor	Audix D4	Local 8		Clip	Template ships this fader gated — thr –36.2 dB, rel 227 ms, 130–317 Hz sidechain. Left as-is.
9	Overheads	Shure Beta 27 (pair)	Local 9	✓	Boom	STEREO channel — both OH mics on this one fader. Their OH L/R collapse here; fader 10 is the SNARE PL8 return, never an input. –15 dB pads assumed OUT. Windscreens.
■ BASS						

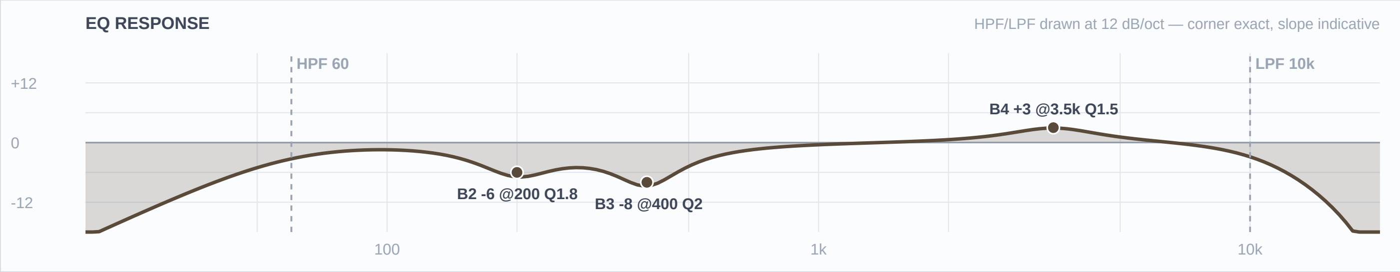
Ch	Instrument	Mic / DI	Patch	48V	Stand	Notes
11	Bass DI	Neve RNDI	Local 11	✓	DI	No bass rig notated. If a 5-string turns up, drop the HPF from 45 to ~35 for the low B.
12	Bass Synth	Radial J48	Local 12	✓	DI	Built for the conservative case that it can play alongside ch 11. If they swap song to song, the HPF can drop to 40 and it can own its own bottom.
■ **GUITAR**						
13	Guitar	XLR line feed (cab sim ON)	Local 13		—	Band's own rig — confirm the feed and its level at load-in. Cab sim confirmed ON.
■ **VOCALS**						
14	Edwin TB	Shure SM58	Local 14		Tall	MUTE BY DEFAULT. Built as mains-capable; CONFIRM whether this is comms/monitors only — if so it needs no EQ and should come out of the record bus.
■ **RHYTHM**						
15	Click	XLR line feed	Local 15		—	MONITORS/IEM ONLY — out of the LR mains, out of any FOH-fed matrix, out of the record bus. Confirm at load-in.
16	Track	XLR line feed	Local 16		—	Band's own playback rig — built as mono; confirm whether it's a stereo pair folded to one channel.
■ **DRUMS**						
17	Conga 1	Audio-Technica PRO 35	Local 17	✓	Clip	AT8538 power module switch assumed FLAT (not the 80 Hz rolloff). If found rolled off, drop the desk HPF to ~90.
18	Conga 2	Audio-Technica PRO 35	Local 18	✓	Clip	AT8538 power module switch assumed FLAT.
19	Bongos	Shure SM57	Local 19		Boom	Between the two drums. Windscreen — gusts to 16 mph.
20	Toys	Shure SM81	Local 20	✓	Boom	Over the toy tray. LF switch assumed FLAT. Windscreen — gusts to 16 mph.
■ **KEYS**						

Ch	Instrument	Mic / DI	Patch	48V	Stand	Notes
21	Pad 1	BSS AR-133 (DI)	Local 21	✓	DI	Run as MONO and panned (Brian's call) — not a stereo pair, so the mismatched DI boxes are a non-issue.
22	Pad 2	Whirlwind IMP (passive DI)	Local 22		DI	Run as MONO and panned. Passive DI — no 48V.
23	Key L	XLR line feed	Local 23		—	Hard-panned left. IDENTICAL EQ to ch 24 — a different curve on L and R makes the image wander.
24	Key R	XLR line feed	Local 24		—	Hard-panned right. IDENTICAL EQ to ch 23.
■ **VOCALS**						
33	Aretha	Shure Beta 58A (Wireless 1)	Local 33		—	House Wireless 1. OVERRIDES the FSQ template's wireless baseline (HPF 184, –18 @ 5k notch, –6.3 @ 335) — RING OUT AT SOUNDCHECK.
34	Heather	Shure Beta 58A (Wireless 2)	Local 34		—	House Wireless 2. Overrides the template's wireless baseline — ring out at soundcheck.
35	Vince	Shure Beta 58A (Wireless 3)	Local 35		—	House Wireless 3. The template's HPF 184 would have removed the bottom of this voice entirely — overridden to 90. Ring out at soundcheck.
36	Markay	Shure Beta 58A (Wireless 4)	Local 36		—	House Wireless 4. Overrides the template's wireless baseline — ring out at soundcheck.

■ DRUMS ■

Ch 1 | Kick In

Mic: Shure Beta 91A



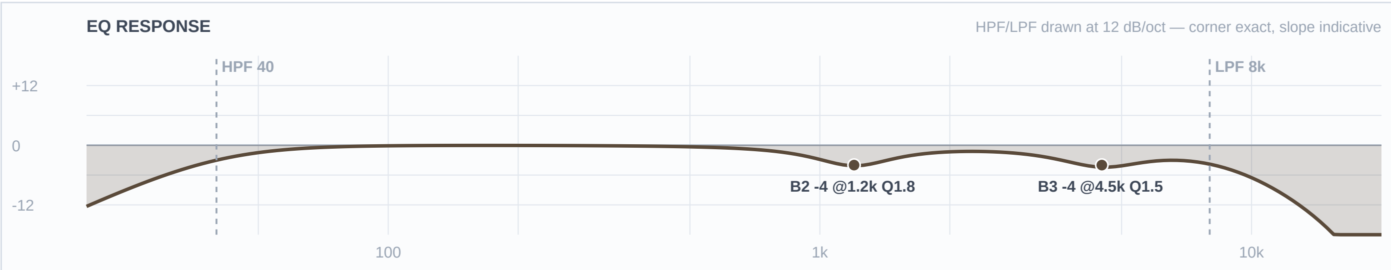
Mic Notes: Shure's own contour switch cuts 7 dB at 400 Hz — that is the manufacturer telling you where this capsule's problem is. Switch assumed FLAT so the desk owns that cut and I can ride it; if it's found engaged, halve the 400 Hz cut. Mix Online calls the 91A's beater snap 'more balanced' than the original 91's — present, not hyped, which is what lets the +3 at 3500 stand as a lift rather than a stack. TWO-MIC PAIR with ch 2: this mic owns the ATTACK/TOP lane (3–6 kHz); the D6 owns the BOTTOM (50–90 Hz). Mono-sum at soundcheck; if thinner than either alone, flip polarity on ch 2, not this one.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	60 Hz	LC	—	—	High-pass
4	3.5 kHz	Bell	+3 dB	1.5	
3	400 Hz	Bell	-8 dB	2	
2	200 Hz	Bell	-6 dB	1.8	
1	—	OFF	—	—	Flat
HC	10 kHz	HC	—	—	Low-pass

Engineer Notes: *The beater channel. HPF at 60 and –6 at 200 deliberately strip the low end off this mic because the D6 outside owns it — stacking two kick mics in the same octave is the classic way to get a big, mushy, undefined kick on a plaza with no room to sort it out. –8 at 400 is the shell box, at the contour switch's own frequency and at FSQ's deeper outdoor depth. The +3 at 3500 is the only boost and it's modest on purpose: an R&B; dance kick wants a defined thud, not a metal click, and the Track channel already has a programmed kick under it.*

Ch 2 | Kick Out

Mic: Audix D6



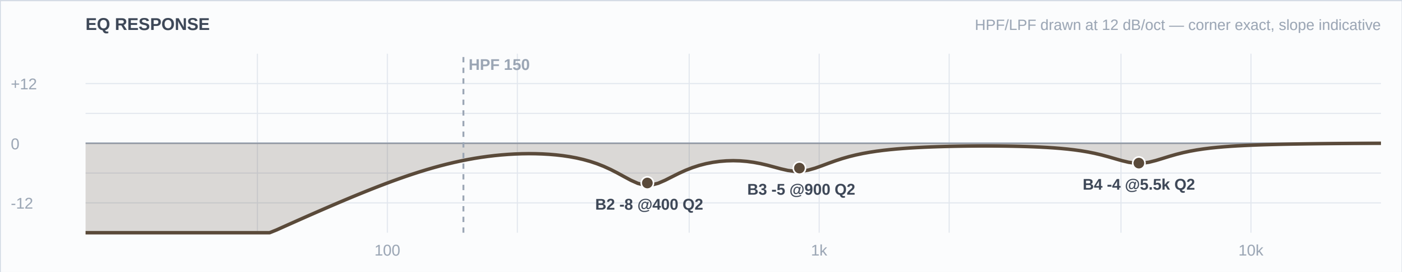
Mic Notes: The most pre-voiced mic on this list: a 15–17 dB scoop centred 700–800 Hz, an LF peak at ~60 Hz (about +5 dB around 40), and HF peaks at 4 kHz and 10 kHz (RecordingHacks/Mix/Tape Op). Everything a kick normally gets boosted, this capsule already boosted — so there are no boosts here at all. TWO-MIC PAIR with ch 1: this mic owns the BOTTOM (its 60 Hz peak is baked in); ch 1 owns the top. Flip polarity on THIS channel if the mono sum is thin.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	40 Hz	LC	—	—	High-pass
4	—	OFF	—	—	Flat
3	4.5 kHz	Bell	-4 dB	1.5	
2	1.2 kHz	Bell	-4 dB	1.8	
1	—	OFF	—	—	Flat
HC	8 kHz	HC	—	—	Low-pass

Engineer Notes: *Almost nothing written, and that's the finding rather than a gap. No low boost — the capsule is already +5 dB down there. No box or mud cut either, even though FSQ's outdoor rule would push one to –8, because the capsule already dug a 17 dB hole at 700–800 and cutting again would just double-dip it. The two moves that remain are both trims of things the capsule bakes in: –4 at 4500 keeps its click out of ch 1's attack lane, and –4 at 1200 pulls its 'definition' peak out of the path of four vocals. The LPF at 8k removes the 10 kHz peak outright.*

Ch 3 | Snare Top

Mic: Audix i5



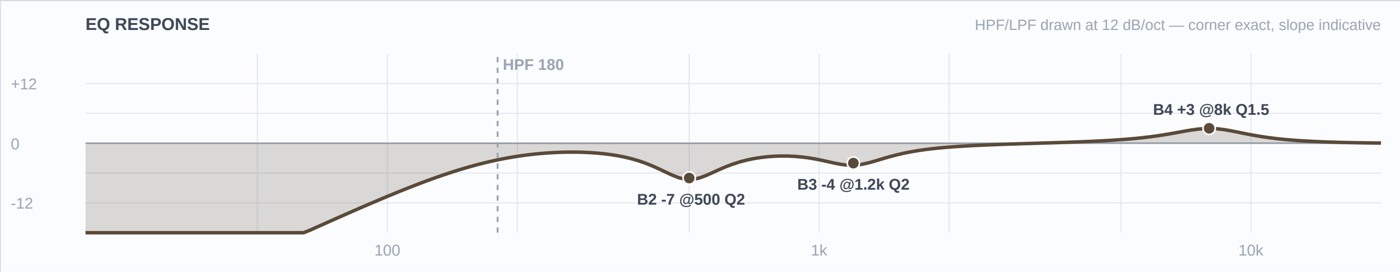
Mic Notes: Locker fork — Beta 98H/C offered, i5 taken (Brian). From the DP8, the same case as the D6, both D2s and the D4, so nothing extra opens for it. RecordingHacks/SOS measure a +5 dB peak at 150 Hz and a +9 dB peak at 5500 Hz — the largest baked peak of any mic on this show. That +9 is exactly why the fork was worth taking AND why this channel gets no crack boost. Being a dynamic it also solves the problem that raised the fork: far less hi-hat bleed than a condenser sitting two feet from an open hat on a windy plaza.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	150 Hz	LC	—	—	High-pass
4	5.5 kHz	Bell	-4 dB	2	
3	900 Hz	Bell	-5 dB	2	
2	400 Hz	Bell	-8 dB	2	
1	—	OFF	—	—	Flat
HC	—	HC	—	—	No LPF

Engineer Notes: *Zero boosts, and the mic is the entire reason. A snare's crack normally gets +3 around 5–7 kHz; this capsule delivers +9 dB at 5500 on its own, and on an outdoor PA with no room to soften it that reads as an ice pick at 60 feet — so the move is –4 and the crack still arrives. Same logic on the bottom: the i5 puts +5 dB at 150 Hz in, so B1 stays flat and the HPF at 150 sits right on that shoulder, keeping the body while cutting kick and floor-tom spill. –8 at 400 is the box at outdoor depth; –5 at 900 keeps the backbeat out of the four vocals' path.*

Ch 4 | Snare 2

Mic: Shure Beta 98H/C



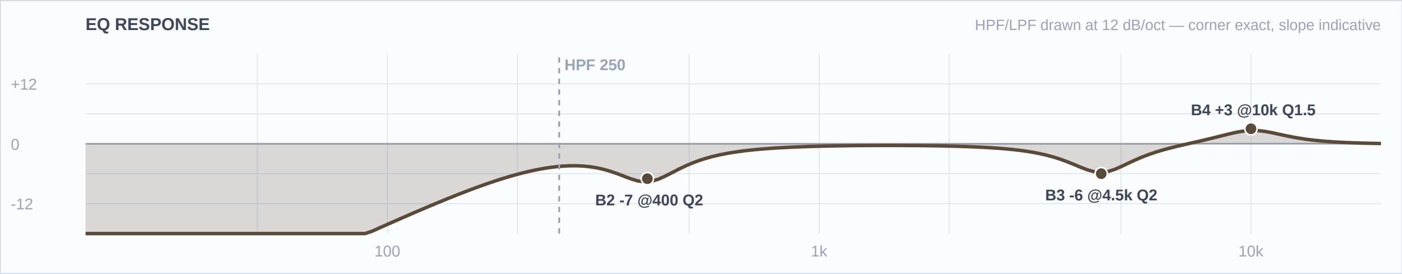
Mic Notes: Locker fork — i5 taken on ch 3; only one in the DP8, so the aux snare keeps the Beta 98H/C. RecordingHacks' measured Beta drum review: a small high-end boost above 8 kHz, more hat bleed than a 57, and enough sub-100 Hz content through the rim to force a high-pass — which contradicts the KB's 'thin lows' note, correctly, because that note describes the mic clipped to a horn bell, not bolted to a snare rim. HPF 180 is that finding applied.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	180 Hz	LC	—	—	High-pass
4	8 kHz	Bell	+3 dB	1.5	
3	1.2 kHz	Bell	-4 dB	2	
2	500 Hz	Bell	-7 dB	2	
1	—	OFF	—	—	Flat
HC	—	HC	—	—	No LPF

Engineer Notes: *The mixed pair turned out better than a matched one. Because this capsule's baked lift starts ABOVE 8 kHz, a +3 AT 8 k is a genuine lift with room under it — the opposite call from ch 3, where the peak sits at 5500 and gets trimmed. That difference does the sectional work for free: crack at 8000 vs 5500, box at 500 vs 400, mid lane at 1200 vs 900, HPF 180 vs 150. A two-snare groove reads as two drums instead of one thick one.*

Ch 5 | Hat

Mic: Electro-Voice N/D 408



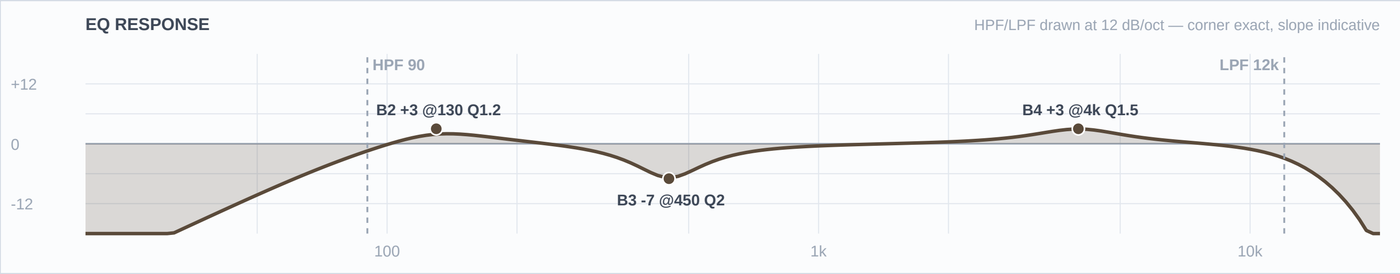
Mic Notes: Locker fork — M1280BHC offered, ND408 kept (Brian). EV's own engineering data sheet (531818-201) shows a broad presence rise centred 4–5 kHz and a close-vs-far response split of 30 Hz–22 kHz against 60 Hz–22 kHz, with the close curve running roughly 10 dB above the far one through 100–300 Hz. The KB's warning is blunt and applies here: 'that upper-mid bite gets spiky on bright sources.' A hi-hat is the definition of a bright source.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	250 Hz	LC	—	—	High-pass
4	10 kHz	Bell	+3 dB	1.5	
3	4.5 kHz	Bell	-6 dB	2	
2	400 Hz	Bell	-7 dB	2	
1	—	OFF	—	—	Flat
HC	—	HC	—	—	No LPF

Engineer Notes: *The –6 at 4500 is load-bearing and it's the price of keeping this mic — the capsule's presence rise sits exactly where hat clank lives, so that band is doing the work the M1280BHC wouldn't have needed doing. Don't dial it out at soundcheck without listening for what comes back. –7 at 400 handles the proximity low-mid pile the data sheet documents. The +3 at 10 k is a genuine lift (the curve is flat-to-rising up there with no baked peak) and it stays modest because humidity climbs 27 points across the set — the top gets MORE present as the night goes on, not less.*

Ch 6 | Rack 1

Mic: Audix D2



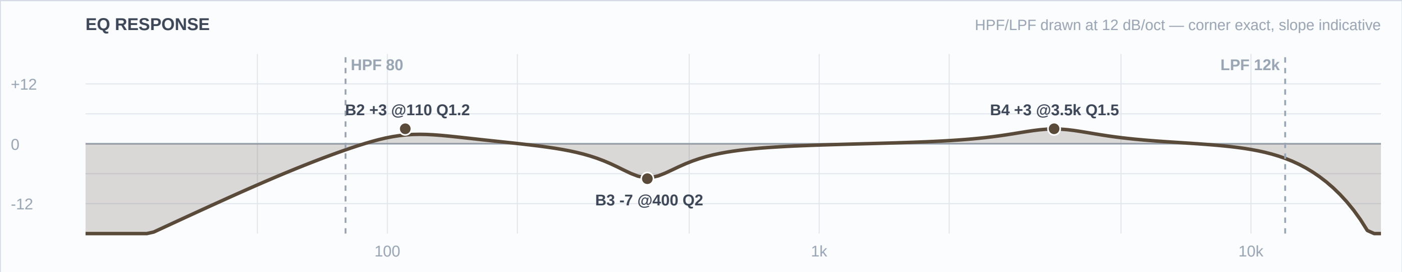
Mic Notes: Audix spec: 68 Hz–18 kHz with a body boost around 150 Hz, a dip between 500 Hz and 1 kHz, a subtle upper-mid lift, and over 30 dB of off-axis rejection. Two of those numbers drive the EQ directly — the body boost is placed BELOW the baked 150 Hz bump so it extends rather than stacks, and B2 is left flat because the capsule already dipped 500 Hz–1 kHz.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	90 Hz	LC	—	—	High-pass
4	4 kHz	Bell	+3 dB	1.5	
3	450 Hz	Bell	-7 dB	2	
2	130 Hz	Bell	+3 dB	1.2	
1	—	OFF	—	—	Flat
HC	12 kHz	HC	—	—	Low-pass

Engineer Notes: *+3 at 130 rather than at 150: the capsule's bump is at 150, so a boost written there would have been the stack this gate exists to catch. B2 stays flat for the same reason in reverse — even with FSQ's deeper-cut rule there is nothing left to cut where the capsule already dipped, so the box cut is placed at 450, below the dip. The +3 at 4000 is stick definition on ground the KB calls 'subtle'. Against ch 7 and ch 8 the three toms descend in both body (130/110/85) and attack (4000/3500/3000), so a fill reads as three drums.*

Ch 7 | Rack 2

Mic: Audix D2



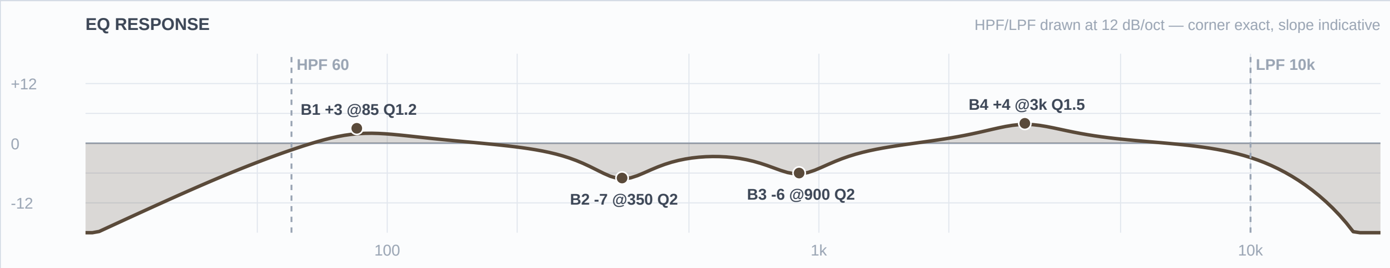
Mic Notes: Same D2 capsule as ch 6 — body boost at ~150 Hz, dip 500 Hz–1 kHz, >30 dB off-axis. The values differ from ch 6 because the drum does, not because the mic does.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	80 Hz	LC	—	—	High-pass
4	3.5 kHz	Bell	+3 dB	1.5	
3	400 Hz	Bell	-7 dB	2	
2	110 Hz	Bell	+3 dB	1.2	
1	—	OFF	—	—	Flat
HC	12 kHz	HC	—	—	Low-pass

Engineer Notes: *The lower rack tom, voiced a step under ch 6 across every band: body at 110 instead of 130, definition at 3500 instead of 4000, box at 400 instead of 450, HPF 80 instead of 90. Both boosts sit below the capsule's baked 150 Hz bump. B2 flat, same capsule reason as ch 6.*

Ch 8 | Floor

Mic: Audix D4



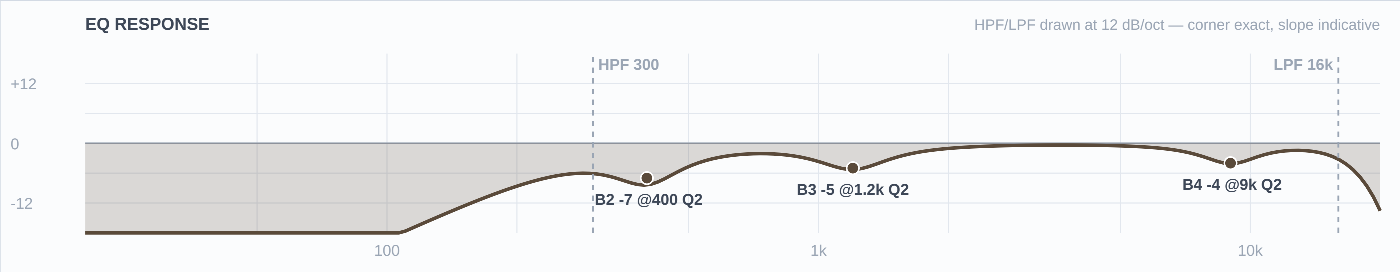
Mic Notes: Audix spec: 40 Hz–18 kHz, hypercardioid, off-axis rejection >30 dB, max SPL >144 dB. The number that matters here is the comparison with its own sibling — the D4 reaches 40 Hz where the D2 stops at 68, which is the whole reason the two mics are on different drums. Neither the spec sheet nor the KB claims any presence peak, and the KB adds the fact that decides B2: the D4's 800 Hz–1 kHz region is SMOOTH, not scooped like the D2's.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	60 Hz	LC	—	—	High-pass
4	3 kHz	Bell	+4 dB	1.5	
3	900 Hz	Bell	-6 dB	2	
2	350 Hz	Bell	-7 dB	2	
1	85 Hz	Bell	+3 dB	1.2	
HC	10 kHz	HC	—	—	Low-pass

Engineer Notes: *The –6 at 900 is the D2/D4 difference made audible: ch 6 and ch 7 leave that band alone because their capsule already dipped it, and this one cuts it at outdoor depth because the D4 doesn't. Same kit, same show, opposite move, and the capsule is the only reason. The +4 at 3000 is the show's largest boost and it's earned — this is the one capsule here with no presence peak and a documented shortage of attack. +3 at 85 sits just above the capsule's own ~80 Hz rise, and clears the bass at 100.*

Ch 9 | Overheads

Mic: Shure Beta 27 (pair)



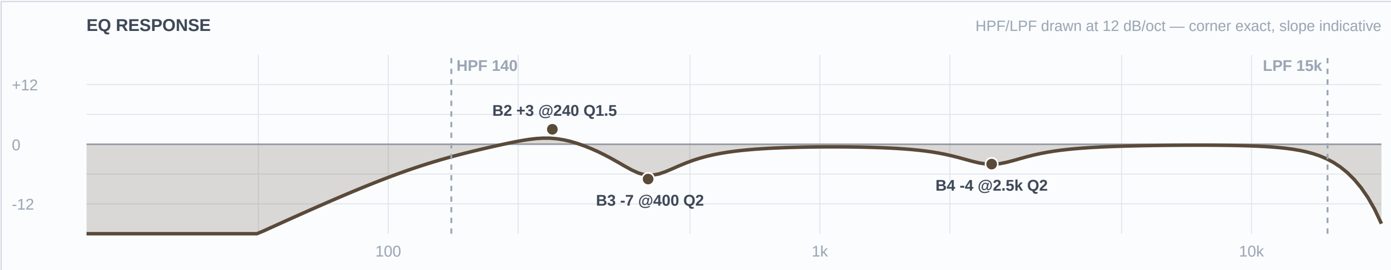
Mic Notes: Nominally flat 60 Hz–3 kHz with two small HF peaks of +2 dB or less at 5.5 kHz and 9 kHz, –3 dB above 15 kHz, switchable –15 dB pad, supercardioid (RecordingHacks). Locker note: the matched Earthworks SR20sp and Audix M1280B pairs are more accurate, but both are cardioid — on an open plaza with a 10,000-person crowd, pattern beats accuracy, so the supercardioid Beta 27s stay. No fork raised.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	300 Hz	LC	—	—	High-pass
4	9 kHz	Bell	-4 dB	2	
3	1.2 kHz	Bell	-5 dB	2	
2	400 Hz	Bell	-7 dB	2	
1	—	OFF	—	—	Flat
HC	16 kHz	HC	—	—	Low-pass

Engineer Notes: *Zero boosts. The two places an overhead normally gets lifted — 5.5 k for stick and 9 k for air — are precisely this capsule's two baked peaks, so the top is a trim: –4 at the 9 k the KB flags as fizz, and the 5.5 k peak left alone because at +2 dB it's doing useful work in a mix this dense. HPF at 300 is far higher than any indoor overhead would run, and it's the right call outdoors with 16 mph gusts on tall booms and close mics already owning the kit. Under a mix with a programmed kit in the Track channel, this fader is glue, not the drum sound.*

Ch 17 | Conga 1

Mic: Audio-Technica PRO 35



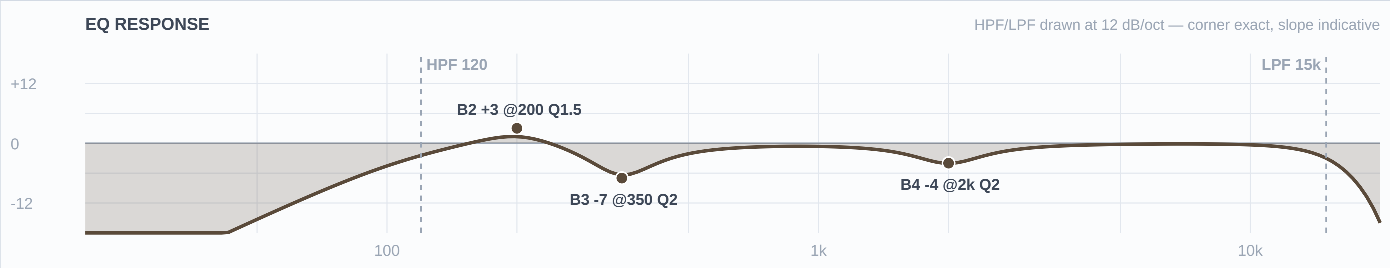
Mic Notes: 50 Hz–15 kHz, 145 dB max SPL, with a switchable 80 Hz roll-off at 18 dB/octave in the AT8538 module (assumed flat). The 15 kHz ceiling is the number that matters — this capsule is done up there, so there is no air to reach for. The KB's warning is the other one: a voiced presence lift that goes 'plasticky in the upper-mid if close', and clip-on means close.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	140 Hz	LC	—	—	High-pass
4	2.5 kHz	Bell	-4 dB	2	
3	400 Hz	Bell	-7 dB	2	
2	240 Hz	Bell	+3 dB	1.5	
1	—	OFF	—	—	Flat
HC	15 kHz	HC	—	—	Low-pass

Engineer Notes: *B4 stays flat because there is genuinely nothing above 15 kHz to lift, and the LPF is set to match the mic rather than pretend otherwise. The –4 at 2500 is the clearest capsule reversal in the show: every percussion chart says boost 2–4 kHz for slap, this capsule already lifts there, so the slap comes from the transient and the desk trims instead. –7 at 400 is the ring harmonic at outdoor depth. +3 at 240 fills the resonance the PRO 35's thin lows don't deliver, held modest because everything below 150 belongs to other channels.*

Ch 18 | Conga 2

Mic: Audio-Technica PRO 35



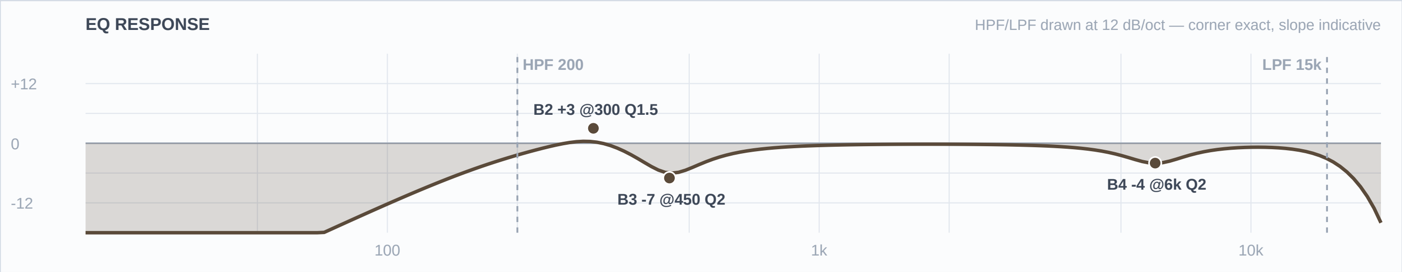
Mic Notes: Same PRO 35 capsule as ch 17 — 15 kHz ceiling, voiced presence lift, thin lows. Same reversal on the slap region for the same reason.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	120 Hz	LC	—	—	High-pass
4	2 kHz	Bell	-4 dB	2	
3	350 Hz	Bell	-7 dB	2	
2	200 Hz	Bell	+3 dB	1.5	
1	—	OFF	—	—	Flat
HC	15 kHz	HC	—	—	Low-pass

Engineer Notes: *The lower drum, voiced a step under ch 17 on every band: tone at 200 vs 240, ring at 350 vs 400, presence trim at 2000 vs 2500, HPF 120 vs 140. The two congas descend together so a pattern reads as two pitches rather than one wide drum. Across the whole percussion chair the tone slots ascend 200 / 240 / 300 (ch 18, 17, 19) with ring cuts at 350 / 400 / 450 to match.*

Ch 19 | Bongos

Mic: Shure SM57



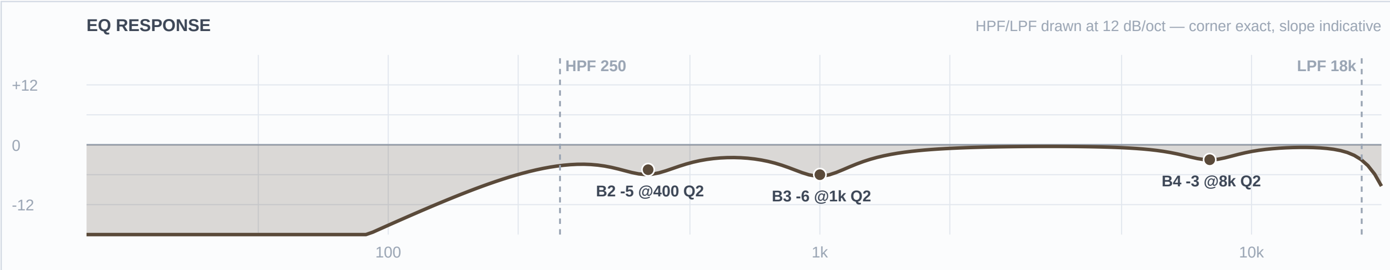
Mic Notes: Shure's own published curve (user guide v3.6, 2025-A): flat from 200 Hz to about 1 kHz, rising to a presence peak of roughly +5 to +6 dB centred 6–7 kHz, about –10 dB at 40 Hz. NOTE — the mic-library row says the peak is at 3–5 kHz; the measured curve is only up ~+2 dB by 3 kHz. Going with the manufacturer's measurement, and the KB correction is offered separately since the 57 is the most-used mic in the locker.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	200 Hz	LC	—	—	High-pass
4	6 kHz	Bell	-4 dB	2	
3	450 Hz	Bell	-7 dB	2	
2	300 Hz	Bell	+3 dB	1.5	
1	—	OFF	—	—	Flat
HC	15 kHz	HC	—	—	Low-pass

Engineer Notes: *Every percussion chart says boost 5 kHz for slap. This capsule already delivers +5 to +6 dB at 6–7 kHz and a bongo is the brightest source on this stage, so on an outdoor PA with nothing to soften it the honest move is –4 and the slap comes from the transient. That trim would have landed in the wrong place using the KB's figure — which is exactly why the correction is worth making. +3 at 300 is a genuine lift into the capsule's flat region, giving the bongos their own tone slot above the two congas at 240 and 200.*

Ch 20 | Toys

Mic: Shure SM81



Mic Notes: 20 Hz–20 kHz genuinely flat, with a three-position LF switch (flat / 6 dB per octave below 100 Hz / 18 dB per octave below 80 Hz), assumed flat so the desk owns the filter. The KB's tendency line for this mic is 'apply template as-is' — it flatters nothing and adds nothing, which is the right property for one mic over a tray of sources that all sound different.

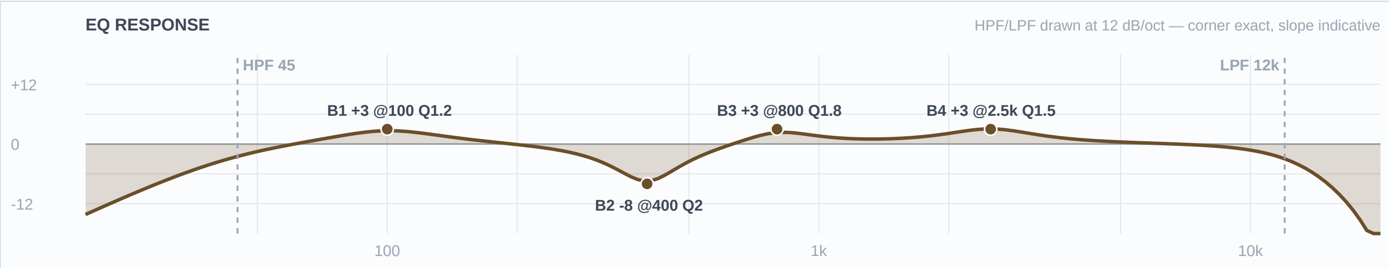
Band	Freq	Type	Gain	Q	Notes / Rationale
LC	250 Hz	LC	—	—	High-pass
4	8 kHz	Bell	-3 dB	2	
3	1 kHz	Bell	-6 dB	2	
2	400 Hz	Bell	-5 dB	2	
1	—	OFF	—	—	Flat
HC	18 kHz	HC	—	—	Low-pass

Engineer Notes: *No boosts at all, and that's the capsule doing its job. A flat mic on sources that are already bright has nothing missing to add — a tambourine through an SM81 arrives with more top than an outdoor PA wants, so the top gets a –3 trim instead of the lift every percussion chart would prescribe. With humidity climbing 50% to 77% across the set that trim will read right at 11pm where a boost would have been unbearable. –6 at 1000 is cowbell and woodblock honk kept out of the four vocals; –5 at 400 is tray and stand box plus stage wash. HPF 250 for the gusts.*

■ BASS ■

Ch 11 | Bass DI

Mic: Neve RNDI



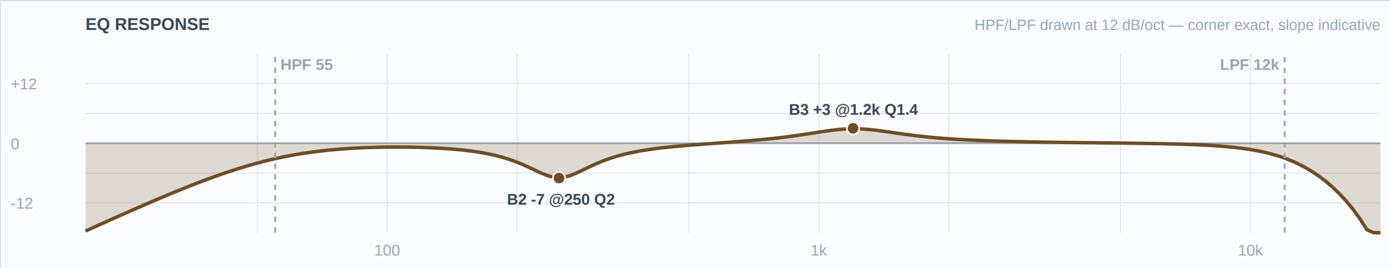
Mic Notes: Active transformer DI — the colour is harmonic warmth and a slight softening of the extreme top, not a frequency peak. No capsule, no baked peak, nothing to gate against, which is why this is the only channel in the show carrying three boosts: it's the one source here with no voicing doing part of the work for it. Instrument anchor: a 4-string's open E is 41 Hz, so HPF 45 keeps the note and loses everything that isn't one.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	45 Hz	LC	—	—	High-pass
4	2.5 kHz	Bell	+3 dB	1.5	
3	800 Hz	Bell	+3 dB	1.8	
2	400 Hz	Bell	-8 dB	2	
1	100 Hz	Bell	+3 dB	1.2	
HC	12 kHz	HC	—	—	Low-pass

***Engineer Notes:** The most important instrument on this stage after the vocals, and it gets slotted rather than just turned up. +3 at 100 sits deliberately above the D6's baked 60 Hz peak — kick owns 60, bass owns 100, and the three gated toms sit at 85/110/130 around it. –8 at 400 is the funk scoop at outdoor depth, deeper than a rock bass would get, because this set is slap-capable and the pop needs the room. +3 at 800 is the pluck definition that makes a line readable at 60 feet on a plaza; +3 at 2500 is the slap articulation.*

Ch 12 | Bass Synth

Mic: Radial J48



Mic Notes: Transparent active DI with tight lows and high headroom — clinical, no colour, which is right for a synth: the source is already a finished sound and doesn't want a transformer's opinion on top. No baked peak of any kind.

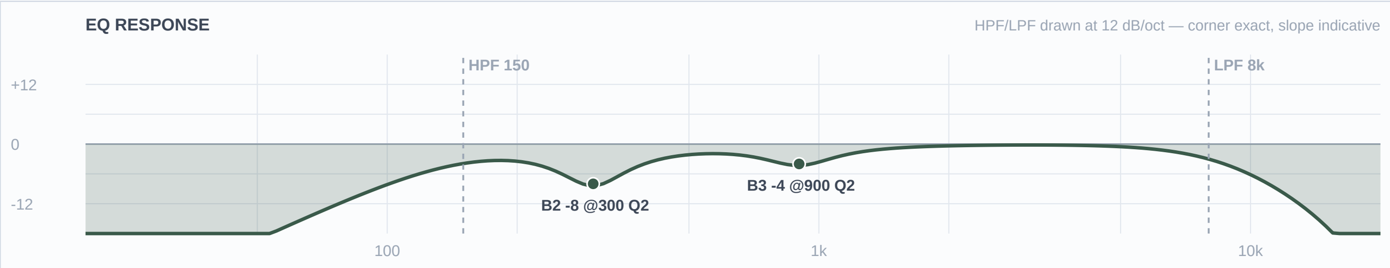
Band	Freq	Type	Gain	Q	Notes / Rationale
LC	55 Hz	LC	—	—	High-pass
4	—	OFF	—	—	Flat
3	1.2 kHz	Bell	+3 dB	1.4	
2	250 Hz	Bell	-7 dB	2	
1	—	OFF	—	—	Flat
HC	12 kHz	HC	—	—	Low-pass

Engineer Notes: *B1 left flat is the deliberate move here. Every instinct says boost a synth bass low; the two-source rule says only one layer owns the fundamental, and ch 11 already does — so this one is high-passed at 55 and earns its place in the mids instead. +3 at 1200 is the band that makes a bass read at distance, and it's placed there specifically because ch 11 puts its definition at 800 and the D6 kick is cutting 1200 — clear air. With a programmed low end already in the Track channel, a third source down at 60 Hz would be one too many.*

■ GUITAR ■

Ch 13 | Guitar

Mic: XLR line feed (cab sim ON)



Mic Notes: No capsule — but a cab-simulated XLR out carries a speaker impulse response with a mic's response already inside it, and that IR does its own presence shaping around 2–4 kHz. So the capsule gate applies here too, one layer further up the chain: a +3 at 3000 was drafted and then withdrawn once Brian confirmed cab sim is ON, because it would have doubled the IR. The LPF at 8k is housekeeping rather than a stand-in for a missing cab.

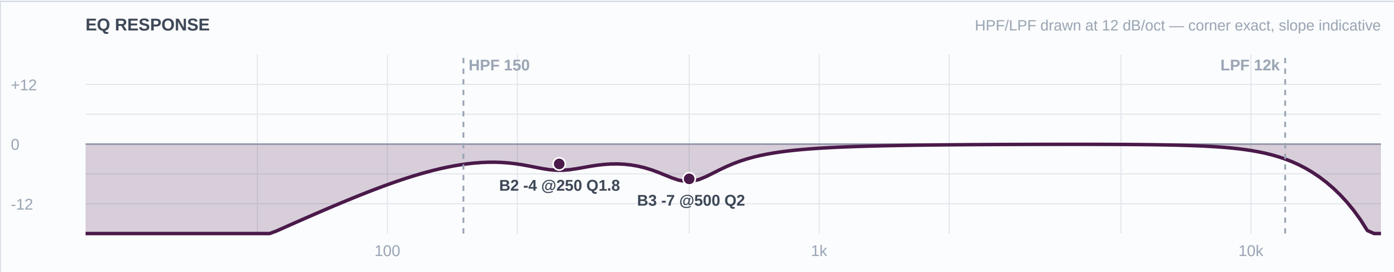
Band	Freq	Type	Gain	Q	Notes / Rationale
LC	150 Hz	LC	—	—	High-pass
4	—	OFF	—	—	Flat
3	900 Hz	Bell	-4 dB	2	
2	300 Hz	Bell	-8 dB	2	
1	—	OFF	—	—	Flat
HC	8 kHz	HC	—	—	Low-pass

***Engineer Notes:** Cuts only, which is the right answer for a finished, already-voiced feed sitting in the middle of a crowded midrange. Funk rhythm guitar wants to be narrow and sharp rather than big — it lives above the bass and below the vocals and its job is rhythmic articulation. HPF at 150 is the top of the researched range (SOS uses 150 on clean funk guitar) and –8 at 300 is the mud cut at outdoor depth, deeper than a rock guitar would get, because mud at 60 feet is what kills a chank. –4 at 900 keeps it clear of the keys at 800 and the vocals above.*

■ VOCALS ■

Ch 14 | Edwin TB

Mic: Shure SM58



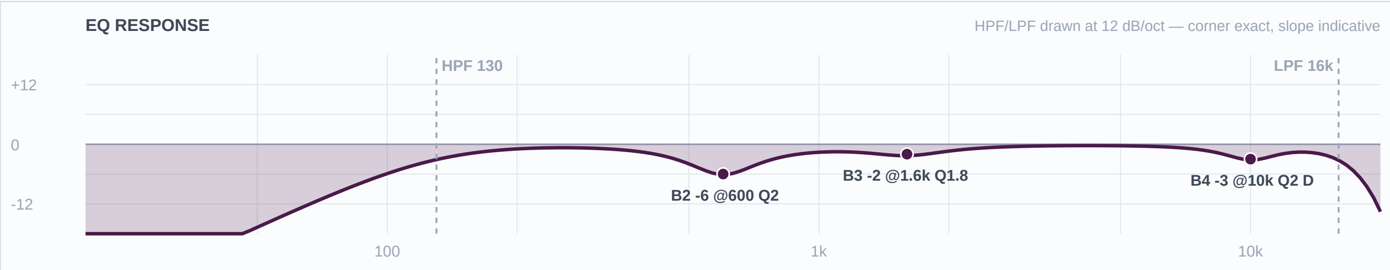
Mic Notes: 50 Hz–15 kHz with a pronounced low rolloff that flattens at 100 Hz and a presence rise running 2 kHz to 6 kHz — Shure designed that rise for speech intelligibility, which is exactly this channel's job. The intelligibility is in the mic; the desk's job is removing what's in the way.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	150 Hz	LC	—	—	High-pass
4	—	OFF	—	—	Flat
3	500 Hz	Bell	-7 dB	2	
2	250 Hz	Bell	-4 dB	1.8	
1	—	OFF	—	—	Flat
HC	12 kHz	HC	—	—	Low-pass

Engineer Notes: *No boosts, for two reasons that both hold independently. The vocals rule is cuts-only in every genre and a talkback is a vocal channel by that rule; and the capsule gate would have blocked the obvious 3–4 kHz intelligibility lift anyway, since this mic already rises across 2–6 kHz. –7 at 500 is the 58's known box at outdoor depth, –4 at 250 handles the proximity chest of a mic that will get eaten. This plus the four wireless are the five open mics that set gain-before-feedback for the whole show.*

Ch 33 | Aretha

Mic: Shure Beta 58A (Wireless 1)



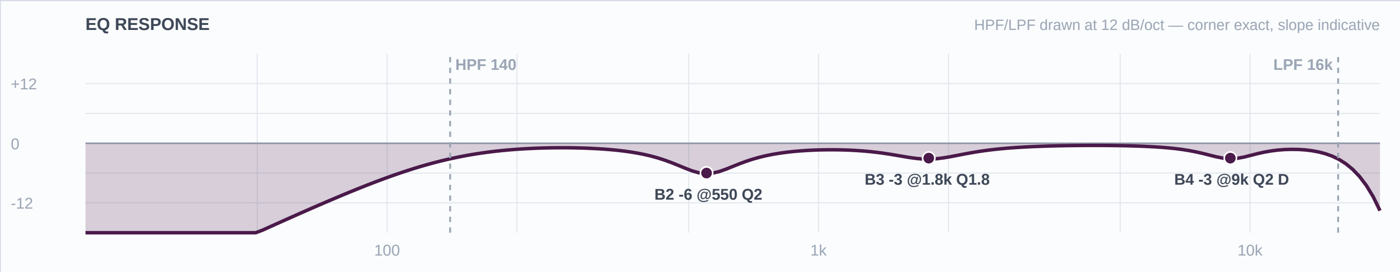
Mic Notes: Beta 58A — two baked presence peaks at 4 kHz and 10 kHz, a pronounced rise through 4–9 kHz that is significantly more aggressive than an SM58's, and a gradually falling bass response below 500 Hz. True supercardioid across the range, which is the gain-before-feedback advantage that makes it the right capsule for four open handhelds on a plaza. COMP: Mustard Purple (Optical), 3:1, attack 10 ms, release 150 ms, threshold for 3–5 dB GR on the belt and none on the verse — documented, not patched.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	130 Hz	LC	—	—	High-pass
4	10 kHz	Bell	-3 dB	2	DYNAMIC: thr=-18dB atk=3ms rel=80ms — bites only on peaks
3	1.6 kHz	Bell	-2 dB	1.8	
2	600 Hz	Bell	-6 dB	2	
1	—	OFF	—	—	Flat
HC	16 kHz	HC	—	—	Low-pass

Engineer Notes: *She's the lead and the MC, so this channel carries speech between songs as well as singing, and it gets the lightest hand of the four. Cuts only — a presence lift at 4k or an air lift at 10k would be stacking the capsule's own two peaks on a trained, loud singer. The de-ess is DYNAMIC rather than static: the 10 kHz peak only misbehaves on sibilants, and with humidity climbing 27 points across the set a fixed value set at 6pm would be wrong by 11. –6 at 600 is the 58A's box at outdoor depth, for intelligibility over a 10,000-person crowd. –2 at 1600 is a light nasal trim, light because she's the voice the show sits on.*

Ch 34 | Heather

Mic: Shure Beta 58A (Wireless 2)



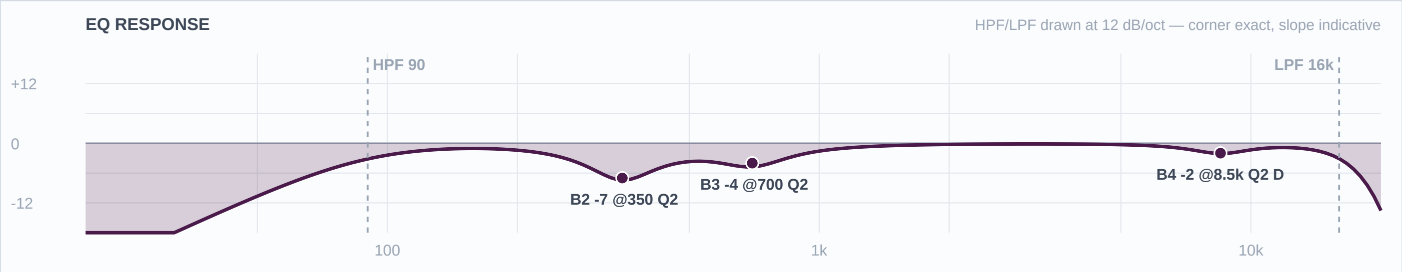
Mic Notes: Same Beta 58A capsule and the same two baked peaks at 4 k and 10 k. The values differ from ch 33 because the voice does, not the mic. COMP: Mustard Purple (Optical), 3:1, 10 ms / 150 ms, 3–5 dB GR on the belt.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	140 Hz	LC	—	—	High-pass
4	9 kHz	Bell	-3 dB	2	DYNAMIC: thr=-18dB atk=3ms rel=80ms — bites only on peaks
3	1.8 kHz	Bell	-3 dB	1.8	
2	550 Hz	Bell	-6 dB	2	
1	—	OFF	—	—	Flat
HC	16 kHz	HC	—	—	Low-pass

Engineer Notes: *The second female voice, slotted against Aretha rather than copied from her: HPF 140 vs 130, box at 550 vs 600, nasal at 1800 vs 1600, de-ess at 9000 vs 10000. Not one shared value. Cuts only, same capsule-gate reasoning — the 4 kHz peak is left completely alone on this channel because it's what carries a harmony over a dance floor. Dynamic de-ess for the same humidity reason as ch 33.*

Ch 35 | Vince

Mic: Shure Beta 58A (Wireless 3)



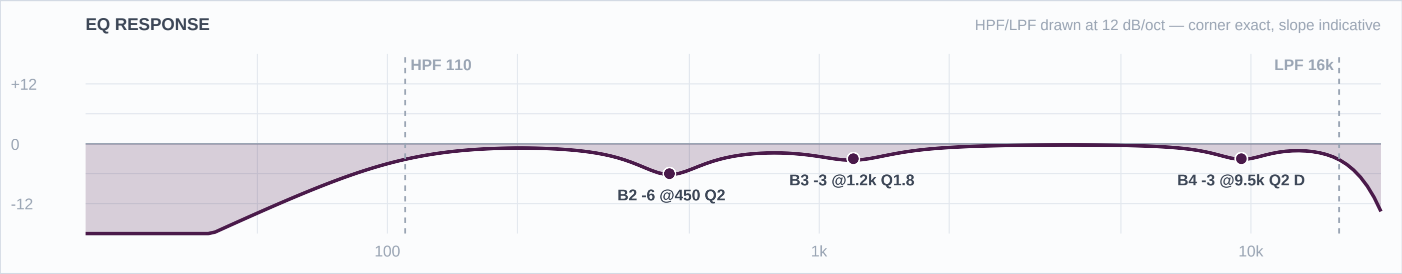
Mic Notes: Same Beta 58A. What differs is the instrument: a bass voice runs E2–E4, roughly 82–330 Hz, with its presence region down at 1–3 kHz — the lowest of the four. That single fact drives every number on this channel. COMP: Mustard Purple (Optical), 3:1, 10 ms / 150 ms — his threshold sits lowest of the four, since a bass voice pushes the most average energy.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	90 Hz	LC	—	—	High-pass
4	8.5 kHz	Bell	-2 dB	2	DYNAMIC: thr=-18dB atk=3ms rel=80ms — bites only on peaks
3	700 Hz	Bell	-4 dB	2	
2	350 Hz	Bell	-7 dB	2	
1	—	OFF	—	—	Flat
HC	16 kHz	HC	—	—	Low-pass

Engineer Notes: *The channel the template would have ruined. Its wireless baseline high-passes at 184 Hz, which sits above this singer's entire lower octave — his fundamentals reach 82 Hz. HPF 90 instead. His chest build-up sits lower than the others' box, so the deep cut is at 350 rather than 550–600, and it's the deepest of the four at –7. The upper-mid cut is the interesting one: at 700 Hz it sits deliberately BELOW his 1–3 kHz presence region, protecting the intelligibility a bass voice can least afford to lose — where the other three are cut inside the 1.2–1.8 kHz nasal zone because their presence sits higher. Lightest de-ess of the four; a bass voice carries less sibilant energy.*

Ch 36 | Markay

Mic: Shure Beta 58A (Wireless 4)



Mic Notes: Same Beta 58A. Tenor range, C3–C5 or roughly 130–523 Hz, presence region 2–5 kHz — so his filter sits between Vince's and the women's, and so does everything else on the channel. COMP: Mustard Purple (Optical), 3:1, 10 ms / 150 ms, 3–5 dB GR on the belt.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	110 Hz	LC	—	—	High-pass
4	9.5 kHz	Bell	-3 dB	2	DYNAMIC: thr=-18dB atk=3ms rel=80ms — bites only on peaks
3	1.2 kHz	Bell	-3 dB	1.8	
2	450 Hz	Bell	-6 dB	2	
1	—	OFF	—	—	Flat
HC	16 kHz	HC	—	—	Low-pass

Engineer Notes: *The upper-range male, and he sits exactly between Vince and the two women on every band: HPF 110 (against 90 and 130/140), box at 450 (against 350 and 550/600), upper-mid at 1200 (against 700 and 1600/1800), de-ess at 9500. Cuts only, and the 4 kHz peak untouched — on the high Usher-style leads that peak is doing the work of getting him over the band. Dynamic de-ess so the top stays right as the humidity climbs through the night.*

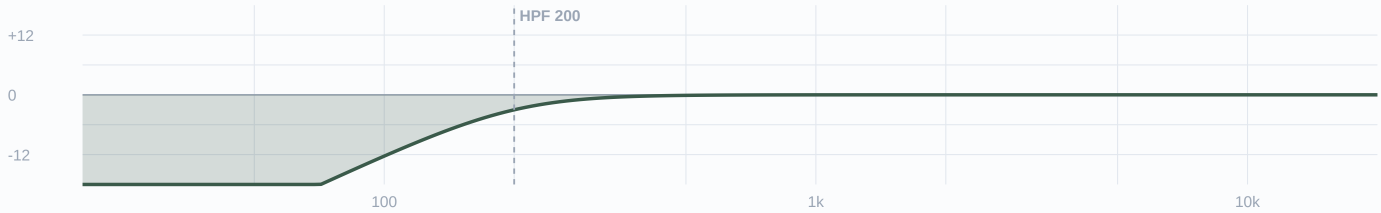
■ RHYTHM ■

Ch 15 | Click

Mic: XLR line feed

EQ RESPONSE

HPF/LPF drawn at 12 dB/oct — corner exact, slope indicative



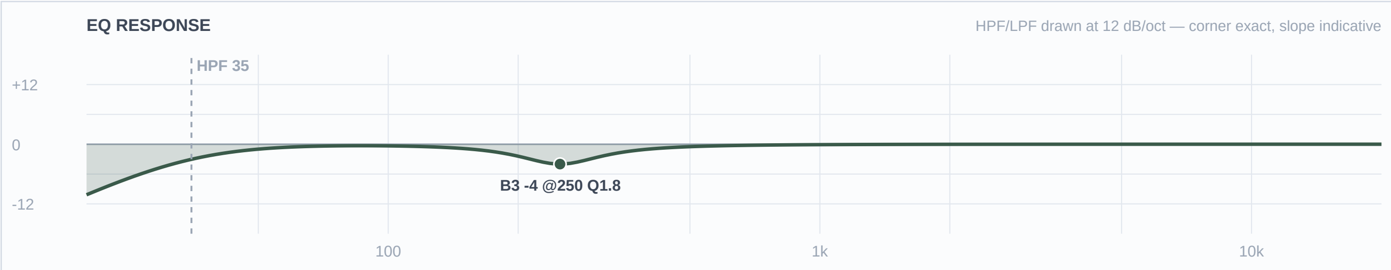
Mic Notes: No capsule and no EQ requirement — a synthesised click is already exactly what it needs to be. The HPF at 200 is housekeeping on the cable run, nothing more.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	200 Hz	LC	—	—	High-pass
4	—	OFF	—	—	Flat
3	—	OFF	—	—	Flat
2	—	OFF	—	—	Flat
1	—	OFF	—	—	Flat
HC	—	HC	—	—	No LPF

Engineer Notes: *The only channel in this show with no EQ at all, deliberately. Its entire requirement is routing, not tone: it exists so the band can hear it in wedges and IEMs and it must never reach the PA or the multitrack. If this one ends up in the mains, no EQ decision on any other channel will matter.*

Ch 16 | Track

Mic: XLR line feed



Mic Notes: Mastered audio — already EQ'd, compressed and limited by whoever produced it, usually to commercial loudness. The most finished signal arriving at this desk, and the one that carries the horn and string parts this 24-channel input list doesn't have.

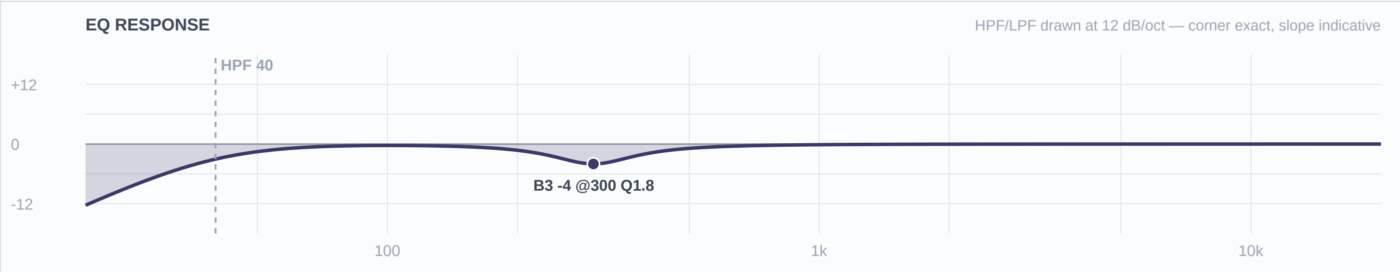
Band	Freq	Type	Gain	Q	Notes / Rationale
LC	35 Hz	LC	—	—	High-pass
4	—	OFF	—	—	Flat
3	250 Hz	Bell	-4 dB	1.8	
2	—	OFF	—	—	Flat
1	—	OFF	—	—	Flat
HC	—	HC	—	—	No LPF

***Engineer Notes:** One band, and that's correct rather than lazy. The gate question for finished audio is whether someone already made this decision, and for a mastered track the answer is yes at every frequency — shaping it means shaping the reference the rest of the mix is being matched to. The single –4 at 250 addresses a problem the producer couldn't have known about: this plaza, with a live kick, bass, bass synth and 808 pads stacked on top. HPF 35 is the lowest filter in the show on purpose — the track's sub content is the point of it.*

■ KEYS ■

Ch 21 | Pad 1

Mic: BSS AR-133 (DI)



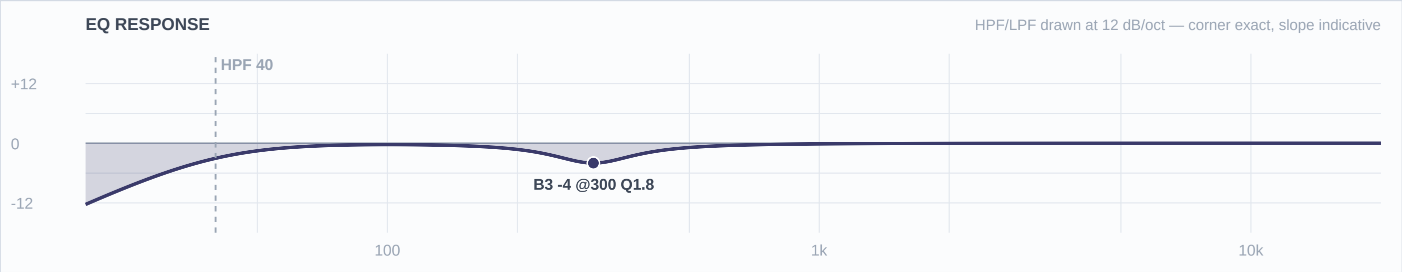
Mic Notes: An SPD-SX carries 21 master effects and 20 kit effects including onboard EQ and compression, and its samples are finished 48 kHz/16-bit audio (Roland). So this signal has already been EQ'd twice before it reaches the desk — once by whoever produced the sample, once by the drummer inside the unit.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	40 Hz	LC	—	—	High-pass
4	—	OFF	—	—	Flat
3	300 Hz	Bell	-4 dB	1.8	
2	—	OFF	—	—	Flat
1	—	OFF	—	—	Flat
HC	—	HC	—	—	No LPF

Engineer Notes: *Almost nothing written, and that's the finding. Two boosts and a deep cut on top of a signal that has already had two opinions applied would be the third. The one moderate 300 Hz trim exists because the plaza stacks low-mids, which is a venue problem the sample's producer couldn't have solved. HPF 40 keeps the sub content the samples were built with — if the pads are firing 808s that land on the D6's baked 60 Hz peak, the fix is the fader and the arrangement, not EQ.*

Ch 22 | Pad 2

Mic: Whirlwind IMP (passive DI)



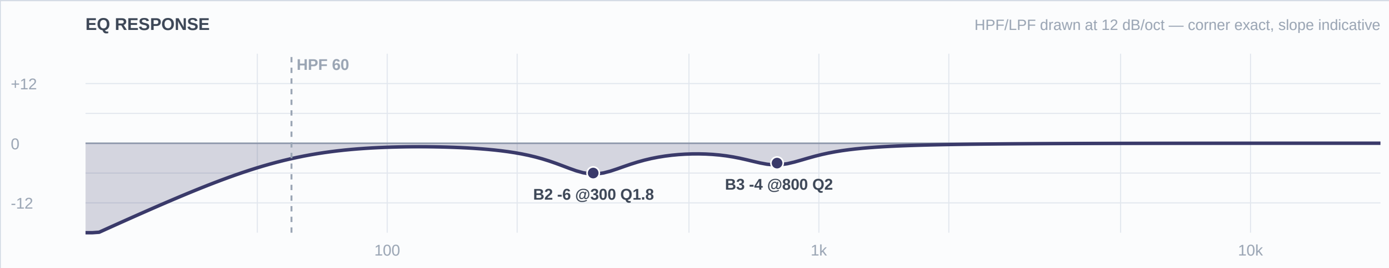
Mic Notes: Passive DI, neutral and honest with a slight HF/level loss into low-Z. Same finished-audio source as ch 21 — the separation between these two channels is pan, not EQ.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	40 Hz	LC	—	—	High-pass
4	—	OFF	—	—	Flat
3	300 Hz	Bell	-4 dB	1.8	
2	—	OFF	—	—	Flat
1	—	OFF	—	—	Flat
HC	—	HC	—	—	No LPF

Engineer Notes: *Identical treatment to ch 21 because it's the same source. Same reasoning: finished audio that has already been decided on twice, one moderate low-mid trim for the outdoor stack, and a low HPF because the sub is what the samples are for.*

Ch 23 | Key L

Mic: XLR line feed



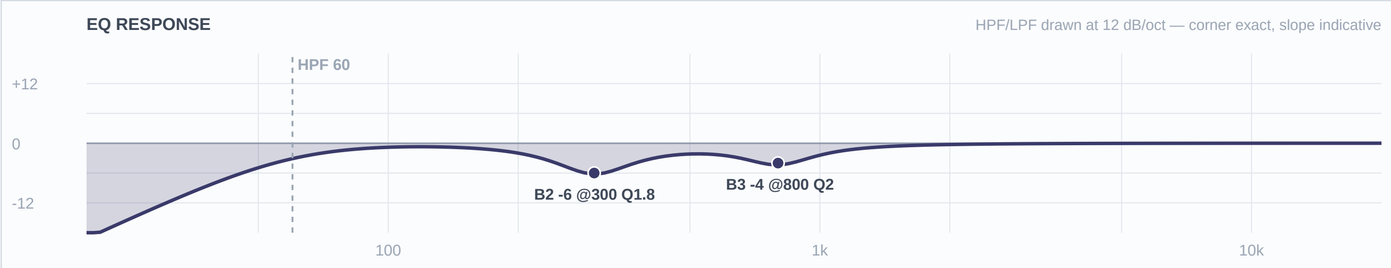
Mic Notes: No capsule — a stage keyboard's XLR out is a finished, already-voiced signal where the patch designer made every tonal decision. Nothing baked in to gate against and nothing missing to restore, which is why this channel is entirely subtractive.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	60 Hz	LC	—	—	High-pass
4	—	OFF	—	—	Flat
3	800 Hz	Bell	-4 dB	2	
2	300 Hz	Bell	-6 dB	1.8	
1	—	OFF	—	—	Flat
HC	—	HC	—	—	No LPF

Engineer Notes: *The widest-ranging chair on this stage: acoustic piano for the ballads, Rhodes for the soul material, B3 for the gospel moments, and — since there are no horn channels on this list — synth brass and strings standing in for the section. One EQ has to serve all of it. The research pulls two ways (cut 300 for Rhodes mud, boost 300–500 for organ weight), and the KB's conservative posture breaks the tie: –6 at 300 rather than the deep end of the outdoor range clears the Rhodes without gutting the organ, and the organ's weight comes back on the fader. The 1–2 kHz window is left deliberately untouched so those synth horn and string parts can speak.*

Ch 24 | Key R

Mic: XLR line feed



Mic Notes: Same finished line feed as ch 23. No boosts on either side — anything the player wants more of, they dial on their own board with far more precision than a 4-band desk EQ gives.

Band	Freq	Type	Gain	Q	Notes / Rationale
LC	60 Hz	LC	—	—	High-pass
4	—	OFF	—	—	Flat
3	800 Hz	Bell	-4 dB	2	
2	300 Hz	Bell	-6 dB	1.8	
1	—	OFF	—	—	Flat
HC	—	HC	—	—	No LPF

Engineer Notes: *Byte-identical to ch 23 by design. Boosting a stage keyboard at FOH is a way of fighting a patch rather than fixing a problem, so both sides stay subtractive: –6 at 300 for the mud stack against bass and guitar, –4 at 800 for the low-mid honk. Keys cut 800 while the guitar cuts 900 and the bass synth boosts 1200 — four midrange sources, four different placements.*